

$$\frac{\delta \ln r_j}{\delta \rho} = \frac{\beta_j(1-v) - 1}{v\rho} < 0 \quad \text{and} \quad \frac{\delta^2 \ln r_j}{\delta \rho \delta \beta_j} = \frac{1-v}{v\rho} > 0.$$

Although we cannot observe β_j , a logical proxy for it is the observed industry level of routine task input in the precomputerization era. We therefore test whether industries that were historically (i.e., precomputer era) intensive in routine tasks adopted computer capital to a greater extent than industries that were not.

- P2. Due to the complementarity between routine and non-routine inputs, a decline in the price of computer capital also raises demand for nonroutine task input. This demand increase is proportionately larger in routine-task-intensive industries:

$$\frac{\delta \ln n_j}{\delta \rho} = \frac{(\beta_j - 1)(1-v)}{v\rho} < 0, \quad \frac{\delta^2 \ln n}{\delta \rho \delta b} = \frac{1-v}{v\rho} > 0.$$

Recall, however, that labor supply to routine tasks declines with ρ . Rising routine task demand must therefore be satisfied with computer capital. Hence, sectors that invest relatively more in computer capital will show a larger rise in nonroutine labor input and a larger decline in routine labor input.

- P3. The previous propositions refer to industry demands. Analogously, we expect that occupations that make relatively larger investments in computer capital will show larger increases in labor input of nonroutine tasks and larger decreases in labor input of routine tasks.

II. EMPIRICAL IMPLEMENTATION

Our analysis requires measures of tasks performed in particular jobs and their changes over time. We draw on information from the Fourth [1977] Edition and Revised Fourth [1991] edition of the U. S. Department of Labor's *Dictionary of Occupational Titles* (DOT). Many of the details of our data construction are provided in the Data Appendix. Here we discuss the most salient features. The U. S. Department of Labor released the first edition of the DOT in 1939 to "furnish public employment offices . . . with information and techniques [to] facilitate proper classification and placement of work seekers" [U. S. Department of Labor 1939;xi, as quoted in Miller et al. 1980]. Although the DOT was

updated four times in the ensuing 60 years [1949, 1965, 1977, and 1991], its structure was little altered. Based upon first-hand observations of workplaces, Department of Labor examiners—using guidelines supplied by the *Handbook for Analyzing Jobs* [U. S. Department of Labor 1972]—evaluate more than 12,000 highly detailed occupations along 44 objective and subjective dimensions, including training times, physical demands and required worker aptitudes, temperaments, and interests.¹¹

Our DOT data are based on an aggregation of these detailed occupations into three-digit Census Occupation Codes (COC), of which there are approximately 450. We append DOT occupation characteristics to the Census Integrated Public Micro Samples [IPUMS, Ruggles and Sobeck 1997] one percent extracts for 1960, 1970, 1980, and 1990, and to CPS Merged Outgoing Rotation Group (MORG) files for 1980, 1990, and 1998. We use all observations for noninstitutionalized, employed workers, ages 18 to 64. For the industry analysis, these individual worker observations are aggregated to the level of 140 consistent Census industries spanning all sectors of the economy in each year of the sample. All analyses are performed using full-time equivalent hours (FTEs) of labor supply as weights. The latter is the product of the individual Census or CPS sampling weight, times hours of work in the sample reference week and, for Census samples, weeks of work in the previous year.

We exploit two sources of variation for measuring changing job task requirements. The first consists of changes over time in the occupational distribution of employment, holding constant task content within occupations at the DOT 1977 level. We refer to cross-occupation employment changes as “extensive” margin shifts, which we can measure consistently over the period 1960 to 1998. This variation does not, however, account for changes in task content within occupations [Levy and Murnane 1996], which we label the “intensive” margin. To measure intensive margin shifts, we analyze changes in task content measures within occupations over the period 1977 to 1991, using occupations matched between the Fourth Edition and Revised Fourth Edition of the DOT.

Although the DOT contains the best time-series on job task

11. The Department of Labor's recent successor to the DOT, O*NET, provides potentially more up-to-date information but is not suitable for time-series analysis.

requirements for detailed occupations in the United States, it also has well-known limitations [Miller et al. 1980]. These include limited sampling of occupations (particularly in the service sector), imprecise definitions of measured constructs, and omission of important job skills. These shortcomings are likely to reduce the precision of our analysis.¹²

II.A. Selecting Measures of Routine and Nonroutine Tasks

To identify variables that best approximate our task constructs, we reduced the DOT variables to a relevant subset using DOT textual definitions, means of DOT measures by occupation from 1970, and detailed examples of DOT evaluations from the *Handbook for Analyzing Jobs*. We selected two variables to measure nonroutine cognitive tasks, one to capture interactive, communication, and managerial skills and the other to capture analytic reasoning skills. The first codes the extent to which occupations involve Direction, Control, and Planning of activities (DCP). It takes on consistently high values in occupations requiring managerial and interpersonal tasks. The second variable, GED-MATH, measures quantitative reasoning requirements. For routine cognitive tasks, we employ the variable STS, which measures adaptability to work requiring Set limits, Tolerances, or Standards. As a measure of routine manual activity, we selected the variable FINGDEX, an abbreviation for Finger Dexterity. To measure nonroutine manual task requirements, we employ the variable EYEHAND, an abbreviation for Eye-Hand-Foot coordination. Definitions and example tasks from the *Handbook for Analyzing Jobs* are provided in Appendix 1.

A limitation of the DOT variables is that they do not have a natural scale and, moreover, cannot confidently be treated as cardinal. To address this limitation, we transformed the DOT measures into percentile values corresponding to their rank in the 1960 distribution of task input. We choose 1960 as the base period for this standardization because it should primarily reflect the distribution of tasks prior to the computer era. Consequently, all of our outcome measures may be interpreted as levels or

12. Researchers who have used the DOT to analyze changing job skill requirements include Rumberger [1981], Spenner [1983, 1990], Howell and Wolff [1991], Wolff [1996, 2002], Handel [2000], and Ingram and Neumann [2000]. What is unique to our work is the focus on routine and nonroutine tasks, and the joint analyses of the effect of computerization on task changes between and within occupations.

changes in task input relative to the 1960 task distribution, measured in "centiles."¹³

II.B. A Predictive Test

As an initial check on our data and conceptual framework, we test the first proposition of our theoretical model: industries historically intensive in routine tasks should have adopted computer capital relatively rapidly as its price fell. To operationalize this test, we form an index of industry-level routine task intensity during the precomputer era. Using the 1960 Census data paired to our selected DOT task measures, we calculate the percentage share of routine task input in industry j 's total task input as, $\text{Routine Task Share}_{j,1960} = 100 \times r_{j,1960} / (r_{j,1960} + n_{j,1960})$, where all task measures are standardized with equal mean and variance. The numerator of this index is the sum of industry routine cognitive and routine manual task inputs, while the denominator is the sum of all five task inputs: routine cognitive and manual; nonroutine analytic, interactive, and manual. This index, which has mean 40.0 and standard deviation 5.0, should roughly correspond to $(1 - \beta_j)$ in our model.

To proxy computer adoption after 1960, we use the Current Population Survey to calculate industries' percentile rank of computer use in 1997. Although we do not have a measure of industry computer use in 1960, this was likely close to zero in all cases. Consequently, the 1997 measure should closely reflect post-1960 computer adoption.

We fit the following equation:

$$(12) \quad \text{Computer adoption}_{j,1960-1997} = -24.56 + 1.85 \times \text{Routine Task Share}_{j,1960} \\ (19.18) \quad (0.48) \quad (n = 140, R^2 = 0.10).$$

The point estimate of 1.85 (standard error 0.48) for the routine task share variable confirms that an industry's routine task intensity in 1960 is strongly predictive of its subsequent computer adoption. Comparing two industries that in 1960 were 10 percentage points (2 standard deviations) apart in routine task input, the model predicts that by 1997, these industries would be 19 percentage points apart in the distribution of computer adop-

13. An earlier version of this paper [Autor, Levy, and Murnane 2001] employed raw DOT scores rather than the percentile measures used here. Results were qualitatively identical.